

Brian Serivuth Ko

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EDUCATION

Texas A&M University, College Station, Texas

Doctor of Philosophy in Biomedical Engineering

Anticipated Feb 2026

Dissertation: Enabling Spatial Analyte Monitoring of Chronic Wounds Using Phosphorescence Lifetime

Imaging of Biosensing Hydrogels

GPR: 3.8

Mississippi State University, Starkville, MS

Bachelor of Science in Biological Engineering, Minor in Economics

May 2019

Honors Thesis: Preliminary study on the effects of *Akkermensia muciniphilia* on Akt phosphorylation in diabetic pig models

GPR: 3.94

SKILLS

Software: Proficient in MATLAB, Microsoft Excel; Familiar with LABView, SOLIDWORKS, COMSOL Multiphysics, ImageJ, Origin Pro

Instrumentation: Proficient in:

Particle Size Analysis: Zetasizer Nano ZS (Malvern) NanoSight LM10 Nanoparticle Tracking Analysis

Spectrofluorometry: PTI Quantamaster with phosphorescence lifetime module

UV-Vis Spectrophotometry: Agilent Cary 300

Scanning Electron Microscopy: JEOL JCM-5000 Neoscope

Transmission Electron Microscopy: JEOL 1200 TEM-EX

Sensor validation models: *in vitro* system, *ex vivo*, rodent, porcine, canine

Molecular skills: Familiar with Western blot

PEER REVIEWED PUBLICATIONS PUBLISHED

L. Singh, M. Yeh, L. Hu, **B. Ko**, M. McShane, and A. Walsh, "A Novel Computational Microscopy Technique for Luminescence Lifetime Imaging," 2025.

R. Pradhan, D. Chimene, **B. S. Ko**, A. Goncharov, A. Ozcan, and M. J. McShane, "Insertable Biomaterial-Based Multianalyte Barcode Sensor toward Continuous Monitoring of Glucose and Oxygen," *ACS Sensors*, vol. 9, no. 11, pp. 6060-6070, 2024/11/22 2024, doi: 10.1021/acssensors.4c01926.

A. Goncharov, Z. Gorocs, R. Pradhan, **B. Ko**, *et al.*, "Insertable Glucose Sensor Using a Compact and Cost-Effective Phosphorescence Lifetime Imager and Machine Learning," *ACS Nano*, vol. 18, no. 34, pp. 23365-23379, 2024/08/27 2024, doi: 10.1021/acsnano.4c06527.

D. Chimene, W. Saleem, N. Longbottom, **B. Ko**, *et al.*, "Long-Term Evaluation of Inserted Nanocomposite Hydrogel-Based Phosphorescent Oxygen Biosensors: Evolution of Local Tissue Oxygen Levels and Foreign Body Response," *ACS Applied Bio Materials*, vol. 7, no. 6, pp. 3964-3980, 2024.

D. Chimene, K. M. K. Queener, **B. S. Ko**, M. McShane, and M. Daniele, "Insertable Biosensors: Combining Implanted Sensing Materials with Wearable Monitors," *Annual Review of Biomedical Engineering*, vol. 26, no. 1, p. null, 2024, doi: 10.1146/annurev-bioeng-110222-101045.

B. Ko, A. T. Zavareh, and M. J. McShane, "In Vitro System for Evaluation of Biosensors in Controlled Dynamic Environmental Conditions," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1-12, 2023.

P. Dong, K. A. Singh, A. M. Soltes, **B. S. Ko**, *et al.*, "Silicone-containing thermoresponsive membranes to form an optical glucose biosensor," *Journal of Materials Chemistry B*, 10.1039/D2TB01192A vol. 10, no. 32, pp. 6118-6132, 2022, doi: 10.1039/D2TB01192A.

P. Dong, **B. S. Ko**, K. A. Lomeli, E. C. Clark, M. J. McShane, and M. A. Grunlan, "A Glucose Biosensor Based on Phosphorescence Lifetime Sensing and a Thermoresponsive Membrane," *Macromolecular Rapid Communications*, vol. 43, no. 9, p. 2100902, 2022, doi: <https://doi.org/10.1002/marc.202100902>.

A. T. Zavareh, **B. Ko**, J. Roberts, S. Elahi, and M. J. McShane, "A Versatile Multichannel Instrument for Measurement of Ratiometric Fluorescence Intensity and Phosphorescence Lifetime," (in eng), *IEEE Access*, vol. 9, pp. 103835-103849, 2021, doi: 10.1109/access.2021.3098777.

RESEARCH EXPERIENCE

Department of Biomedical Engineering, Texas A&M University

Aug 2019 – Present

Graduate Research Assistant, Advisor: Prof. Michael McShane

Enabling Spatial Analyte Monitoring of Chronic Wounds Using Phosphorescence Lifetime Imaging of Biosensing Hydrogels

- Synthesis of oxygen-insensitive phosphorescent nanoparticles for more accurate enzymatic glucose sensing
- Development of an *in vitro* validation device for spatial analyte monitoring using phosphorescence lifetime
- Collaboration with academic and industry partners for phosphorescence lifetime sensing
- Conducted rodent and porcine model studies of oxygen and glucose sensor biocompatibility and performance
- Mentored undergraduate and graduate researchers
- Onboarded incoming graduate students

Department of Food Science, Nutrition, and Health Promotion, Mississippi State University

Sept 2018 – May 2019

Undergraduate Research Assistant, Advisor: Prof. Wen-Hsing Cheng

- Studied effects of probiotic *A. muciniphila* on glycolysis metabolic pathway and development of pre-diabetes and diabetes in various porcine organ tissue samples

PRESENTATIONS

CONFERENCE PRESENTATIONS

B. S. Ko, R. Pradhan, and M. J. McShane, "Phosphorescence Lifetime Imaging for Monitoring an Insertable Hydrogel Multianalyte Sensor," in *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, July 2024.

POSTER PRESENTATIONS

B. Ko et al., "Spatially multiplexed glucose, oxygen, and temperature sensing through synthetic tissue phantoms," in *Optical Diagnostics and Sensing XXV: Toward Point-of-Care Diagnostics*, 2025, vol. 13316: BiOS at Photonics West, Jan 2025.

B. Ko and M. J. McShane, "In vitro evaluation of an Insertable Hydrogel Multianalyte Sensor," in *BMES Annual Meeting*, Oct 2023.

TEACHING EXPERIENCE

Department of Chemistry, Texas A&M University

Teaching Assistant, General Chemistry for Engineers

Aug 2025 – Dec 2025

- Facilitated weekly laboratory sessions for ~100 freshmen students through lab experiments covering fundamental chemistry principles and lab techniques

- Provided one-on-one and group instruction to deliver complex concepts and ensure student comprehension of the material
- Graded assignments and gave constructive feedback in a timely manner

Department of Biomedical Engineering, Texas A&M University

Instructional Service Assistant, Bioinstrumentation Lab

Aug 2020 – Dec 2021

- Facilitated weekly laboratory sessions for ~40 students in hybrid (in-person and remote) format to cover basic electronics, signal acquisition, and data analysis
- Provided hands-on assistance and troubleshooting for experimental setups
- Encouraged a supportive learning environment by explaining complex concepts and answering student questions during lab sessions and virtual office hours
- Graded lab reports and provided constructive feedback to aid student understanding

LEADERSHIP

SPIE Student Chapter at Texas A&M University

President, Vice-President, Outreach Coordinator

Sept 2019 - Dec 2023

- Refreshed existing and developed new outreach activities for K12 outreach in Bryan/College Station, wider Brazos Valley area, and greater Houston
- Organized and led bi-weekly technical and professional development seminars
- Established and facilitated correspondence project between 25 undergraduate/graduate students at TAMU and 25 middle school students in South Carolina
- Awarded a \$5,000 SPIE grant to develop and ship out educational kits pertaining to optics

PROFESSIONAL SOCIETIES

- IEEE, Student Member
- SPIE, Student Member

Sept 2019 - present

Sept 2019 - present

AWARDS

TAMU Office of Graduate and Professional Studies Diversity Excellence Fellowship

Aug 2019